

DRAFT: Optimal Harvesting in a Stochastic Predator–Prey
System:
Analytical Characterization and Numerical Resolution

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Abstract

We study the problem of optimally harvesting two interacting species—one a predator of the other—modeled using a stochastic Lotka–Volterra system with harvesting. The objective is to maximize total expected profit from fishing both species over a finite time horizon, accounting for nonlinear extraction costs and environmental stochasticity. We derive the necessary conditions for optimality using the stochastic maximum principle, in cases where the diffusion is and is not dependent on the control. Where analytical solutions are intractable, we employ a forward-backward sweep method to compute the optimal harvesting strategy numerically. The paper concludes with simulations and comparisons with linearized and deterministic approximations.

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1 Introduction

Sustainable management of renewable biological resources—especially fisheries—has long been a central challenge in both ecological and economic modeling. Classical approaches have addressed optimal harvesting for single species under deterministic assumptions, often using tools from calculus of variations or optimal control. However, such models neglect important features of real-world ecosystems, including interspecies interactions and environmental uncertainty.

In this paper, we consider a more realistic setting: a predator–prey fishery in which both species are subject to harvesting, and their populations evolve stochastically due to environmental variability. Specifically, we model the ecosystem with a stochastic version of the classical Lotka–Volterra system, incorporating both ecological dynamics and economic incentives. The goal is to determine optimal harvesting strategies for the predator and prey populations that maximize expected profit over a finite time horizon. We account for price differences between species and include nonlinear harvesting costs that increase when populations are small or harvest rates are high.

A novel feature of this study is the comparison between two models of environmental noise: one in which the diffusion term (representing stochastic fluctuations) is independent of the harvesting effort, and another in which the diffusion explicitly depends on the control. The latter captures scenarios where intense harvesting increases ecosystem variability, either through habitat disruption or destabilizing biological interactions.

To solve each problem, we employ the stochastic maximum principle (SMP) to derive necessary conditions for optimality and formulate the associated forward-backward stochastic differential equations (FBSDEs). We then implement the forward-backward sweep method to solve these systems numerically and simulate optimal population and control trajectories under both diffusion assumptions.

This study contributes a comparative analysis of two widely relevant modeling approaches in stochastic ecological control. By quantifying the effects of control-dependent diffusion, we provide insight into when more complex models are justified and how they influence optimal harvesting decisions.

2 Model Formulation

We consider a finite-horizon stochastic control problem in a two-species fishery where one species is a natural predator of the other. Let $x_1(t)$ and $x_2(t)$ denote the population densities of the prey and predator species, respectively, at time $t \in [0, T]$. The baseline dynamics are based on the well-known Lotka–Volterra (LV) model [2, 6], which captures predator-prey interactions through the system:

$$\frac{dx_1}{dt} = \alpha x_1 - \beta x_1 x_2 \quad (1)$$

$$\frac{dx_2}{dt} = -\gamma x_2 + \delta x_1 x_2 \quad (2)$$

where α and γ represent intrinsic growth and death rates of the prey and predator species, respectively, and β , δ represent interaction coefficients. All parameters are assumed positive.

This model assumes that prey reproduce exponentially in the absence of predation and that the interaction rate is proportional to the product of the species populations. However, real ecosystems are subject to:

- **Harvesting by humans**, which depletes population levels.
- **Environmental uncertainty**, which leads to stochastic fluctuations in population dynamics.
- **Economic incentives**, including species prices and harvesting costs.

We now introduce harvesting controls and stochasticity into the model, and consider two formulations:

2.1 Model A: Control-Independent Diffusion

In this formulation, population fluctuations are purely environmental and not directly influenced by harvesting intensity. The population dynamics are governed by:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)] dt + \sigma_1 x_1(t) dW_1(t) \quad (3)$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)] dt + \sigma_2 x_2(t) dW_2(t) \quad (4)$$

where:

- $u_1(t), u_2(t) \geq 0$ are the harvesting rates for prey and predator.
- $W_1(t)$ and $W_2(t)$ are independent standard Wiener processes.
- σ_1, σ_2 quantify the intensity of environmental variability.

2.2 Model B: Control-Dependent Diffusion

Here we allow harvesting intensity to influence environmental variability, modeling the idea that aggressive harvesting may destabilize ecosystems or introduce additional uncertainty. The dynamics become:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)] dt + (\sigma_{10}x_1(t) + \sigma_{11}u_1(t)) dW_1(t) \quad (5)$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)] dt + (\sigma_{20}x_2(t) + \sigma_{22}u_2(t)) dW_2(t) \quad (6)$$

where the terms $\sigma_{11}u_1(t)$ and $\sigma_{22}u_2(t)$ represent additional volatility induced by harvesting activity.

2.3 Objective Functional

For both models, the controller aims to maximize the expected net profit:

$$J(u_1, u_2) = \mathbb{E} \left[\int_0^T \left(F_1 u_1(t) + F_2 u_2(t) - \frac{\gamma_1 u_1(t)^2}{x_1(t)} - \frac{\gamma_2 u_2(t)^2}{x_2(t)} \right) dt \right] \quad (7)$$

where

- $F_1, F_2 > 0$ are the market prices for prey and predator species.
- $\gamma_1, \gamma_2 > 0$ scale the nonlinear harvesting costs, which increase when populations are small or effort is high.

2.4 Admissibility and Constraints

The controls $u_1(t), u_2(t)$ must be non-negative, adapted to the natural filtration of $\{W_1(s), W_2(s)\}_{s \leq t}$, and satisfy integrability conditions ensuring that $J(u_1, u_2)$ is well-defined. State and control variables must satisfy:

$$x_1(t) > 0, \quad x_2(t) > 0, \quad u_1(t) \geq 0, \quad u_2(t) \geq 0, \quad \forall t \in [0, T]. \quad (8)$$

In the next section, we derive the necessary optimality conditions for both models using the stochastic maximum principle. This includes the construction of Hamiltonians, derivation of adjoint equations, and characterization of the optimal controls.

3 Stochastic Maximum Principle and Optimality Conditions

To characterize the optimal harvesting strategies $(u_1^*(t), u_2^*(t))$, we apply the stochastic maximum principle (SMP), which provides necessary conditions for optimality in controlled Itô diffusion systems. We follow the formulation in [7], focusing on the case where the drift depends on the control. In this section, we consider two settings: Model A, in which the diffusion term is independent of the control, and Model B, where it is control-dependent.

Let the net profit rate be denoted by $f(x_1, x_2, u_1, u_2)$ and suppose there is no terminal reward, denoted by $g(x_1(T), x_2(T)) = 0$. The objective is to maximize the expected total profit:

$$J(U) = \mathbb{E} \left[\int_0^T f(X(t), U(t)) dt \right] \quad (9)$$

subject to the stochastic dynamics of the population densities. For generality, we write the system as:

$$dX(t) = b(X(t), U(t)) dt + \Sigma(X(t), U(t)) dW(t), \quad (10)$$

where:

- $X(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$ is the state vector of prey and predator populations.
- $U(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$ is the control vector of harvesting rates.
- $W(t) = \begin{bmatrix} W_1(t) \\ W_2(t) \end{bmatrix}$ is a 2-dimensional standard Brownian motion.
- $b(X, U) = \begin{bmatrix} \alpha x_1 - \beta x_1 x_2 - u_1 \\ -\gamma x_2 + \delta x_1 x_2 - u_2 \end{bmatrix}$ is the drift term, which is the same for both models
- $\Sigma : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^{2 \times 2}$ is the diffusion matrix, which differs between models A and B.

Additionally, the profit function is:

$$f(X, U) = F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2}. \quad (11)$$

The stochastic maximum principle introduces an adjoint process $(P(t), Q(t))$ satisfying a backward stochastic differential equation (BSDE):

$$dP(t) = - \left[\nabla_x b(X, U)^\top P(t) + \sum_{j=1}^2 \nabla_x \sigma^j(X, U)^\top Q_j(t) + \nabla_x f(X, U) \right] dt + Q(t) dW(t), \quad (12)$$

where σ^j denotes the j -th column of the diffusion matrix Σ and $Q(t)$ is a 2×2 matrix whose j -th column is $Q_j(t)$. The bracket is precisely $\nabla_x H$ for the Hamiltonian H defined in (15), so (12) is the familiar $dP = -\nabla_x H dt + Q dW$ with terminal condition $P(T) = \nabla_x g(X(T)) = 0$. In particular all three terms enter with the same sign; the running profit f contributes $+\nabla_x f$, not $-\nabla_x f$. This matters here because f depends on the state only through the cost term $-\gamma_i u_i^2/x_i$, whose derivative $+\gamma_i u_i^2/x_i^2$ is the sole forcing of the adjoint system.

3.1 Model A: Control-Independent Diffusion

In this case, we take $\Sigma(X)$ to be a diagonal matrix:

$$\Sigma(X) = \begin{bmatrix} \sigma_1 x_1 & 0 \\ 0 & \sigma_2 x_2 \end{bmatrix}, \quad \text{with } \sigma_1, \sigma_2 > 0.$$

Let $P(t) = \begin{bmatrix} p_1(t) \\ p_2(t) \end{bmatrix}$ and let $Q(t) = (q_{ij}(t))$ be the associated 2×2 martingale coefficient. Because Σ is diagonal, $\nabla_x \sigma^1 \top Q_1 = (\sigma_1 q_{11}, 0)^\top$ and $\nabla_x \sigma^2 \top Q_2 = (0, \sigma_2 q_{22})^\top$, and likewise $\text{tr}(Q^\top \Sigma) = \sigma_1 x_1 q_{11} + \sigma_2 x_2 q_{22}$. Only the diagonal entries of Q therefore appear in the Hamiltonian, in the adjoint drift, or in the optimality condition. We abbreviate $q_1 := q_{11}$ and $q_2 := q_{22}$. (This is a statement

about which entries are dynamically relevant, not a claim that Q is diagonal: p_1 is driven by both Brownian motions, since its drift depends on x_2 , so q_{12} and q_{21} are generally nonzero. They simply never feed back into the system and need not be computed.) The adjoint equations are then

$$dp_1(t) = - \left[(\alpha - \beta x_2)p_1 + \delta x_2 p_2 + \sigma_1 q_1 + \frac{\gamma_1 u_1^2}{x_1^2} \right] dt + q_1 dW_1(t) + q_{12} dW_2(t), \quad (13)$$

$$dp_2(t) = - \left[-\beta x_1 p_1 + (-\gamma + \delta x_1)p_2 + \sigma_2 q_2 + \frac{\gamma_2 u_2^2}{x_2^2} \right] dt + q_{21} dW_1(t) + q_2 dW_2(t), \quad (14)$$

with terminal conditions $p_1(T) = 0$, $p_2(T) = 0$. The Hamiltonian for this model is:

$$H(X, U, P, Q) = f(X, U) + P^\top b(X, U) + \text{tr}(Q^\top \Sigma(X)) \quad (15)$$

$$\begin{aligned} &= F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2} \\ &\quad + p_1(\alpha x_1 - \beta x_1 x_2 - u_1) + p_2(-\gamma x_2 + \delta x_1 x_2 - u_2) \\ &\quad + q_1 \sigma_1 x_1 + q_2 \sigma_2 x_2 \end{aligned} \quad (16)$$

Since the diffusion is independent of control, the SMP condition for optimality is:

$$H(X^*, U^*, P, Q) = \max_{U \in \mathcal{U}} H(X, U, P, Q). \quad (17)$$

Setting the derivatives $\partial H / \partial u_i = 0$, we find the optimal controls (assuming interior maxima):

$$\frac{\partial H}{\partial u_1} = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 = 0 \quad \Rightarrow \quad u_1^* = \frac{x_1}{2\gamma_1} (F_1 - p_1) \quad (18)$$

$$\frac{\partial H}{\partial u_2} = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 = 0 \quad \Rightarrow \quad u_2^* = \frac{x_2}{2\gamma_2} (F_2 - p_2) \quad (19)$$

These expressions define the optimal harvesting policy in terms of the current population and the adjoint process values. While the adjoint processes $p_1(t), p_2(t)$ are governed by backward stochastic differential equations (BSDEs), the population dynamics in equation (10) define forward stochastic processes. Substituting the optimal controls

$$U^*(t) = \begin{bmatrix} u_1^*(t) \\ u_2^*(t) \end{bmatrix}$$

into the system yields a coupled system of forward-backward stochastic differential equations (FB-SDEs), which can be solved numerically to obtain the optimal harvesting strategy.

3.2 Model B: Control-Dependent Diffusion

In this case, we allow the diffusion coefficients to depend explicitly on the control variables, capturing additional variability due to harvesting intensity. The diffusion matrix $\Sigma(X, U)$ is still diagonal, but now has control-dependent entries:

$$\Sigma(X, U) = \begin{bmatrix} \sigma_{10}x_1 + \sigma_{11}u_1 & 0 \\ 0 & \sigma_{20}x_2 + \sigma_{22}u_2 \end{bmatrix}. \quad (20)$$

Recall the forward dynamics are now:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)] dt + (\sigma_{10}x_1(t) + \sigma_{11}u_1(t)) dW_1(t), \quad (21)$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)] dt + (\sigma_{20}x_2(t) + \sigma_{22}u_2(t)) dW_2(t). \quad (22)$$

We use the same first-order adjoint process $(P(\cdot), Q(\cdot))$ described by (12). For this model, we have

$$dp_1(t) = - \left[(\alpha - \beta x_2)p_1 + \delta x_2 p_2 + \sigma_{10} q_1 + \frac{\gamma_1 u_1^2}{x_1^2} \right] dt + q_1 dW_1(t) + q_{12} dW_2(t), \quad (23)$$

$$dp_2(t) = - \left[-\beta x_1 p_1 + (-\gamma + \delta x_1)p_2 + \sigma_{20} q_2 + \frac{\gamma_2 u_2^2}{x_2^2} \right] dt + q_{21} dW_1(t) + q_2 dW_2(t), \quad (24)$$

Note that only σ_{10} and σ_{20} appear: the control-dependent part of the diffusion does not depend on x , so it drops out of $\nabla_x \sigma^j$.

To account for control-dependent noise, we now introduce a second-order adjoint process, solved by $(R(\cdot), S(\cdot))$. Let:

$$R(t) = \begin{bmatrix} r_{11}(t) & r_{12}(t) \\ r_{21}(t) & r_{22}(t) \end{bmatrix}, \quad S_j(t) = \begin{bmatrix} s_{11}^{(j)}(t) & s_{12}^{(j)}(t) \\ s_{21}^{(j)}(t) & s_{22}^{(j)}(t) \end{bmatrix} \quad \text{for } j = 1, 2.$$

These processes satisfy a second-order backward stochastic differential equation,

$$dR(t) = - \left[b_x^\top R + R b_x + \sum_{j=1}^2 \sigma_x^{j\top} R \sigma_x^j + \sum_{j=1}^2 \left(\sigma_x^{j\top} S_j + S_j \sigma_x^j \right) + H_{xx} \right] dt + \sum_{j=1}^2 S_j(t) dW_j(t), \quad (25)$$

where b_x and σ_x^j denote the Jacobians of the drift and diffusion coefficients with respect to the state variables X , and H is the Hamiltonian defined in (15). The BSDE has the terminal condition:

$$R(T) = -g_{xx}(X^*(T)) = 0.$$

Solving for the dynamics of the second-order adjoint process, we find:

$$\begin{aligned} dR(t) = - \left\{ \begin{bmatrix} \alpha - \beta x_2 & \delta x_2 \\ -\beta x_1 & -\gamma + \delta x_1 \end{bmatrix}^\top R(t) + R(t) \begin{bmatrix} \alpha - \beta x_2 & \delta x_2 \\ -\beta x_1 & -\gamma + \delta x_1 \end{bmatrix} \right. \\ + \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix}^\top R(t) \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix}^\top R(t) \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix} \\ + \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix}^\top S_1(t) + S_1(t) \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix}^\top S_2(t) + S_2(t) \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix} \\ \left. + \begin{bmatrix} -2\gamma_1 \frac{u_1^2}{x_1^3} & \delta p_2 - \beta p_1 \\ \delta p_2 - \beta p_1 & -2\gamma_2 \frac{u_2^2}{x_2^3} \end{bmatrix} \right\} dt + S_1(t) dW_1(t) + S_2(t) dW_2(t), \quad (26) \end{aligned}$$

where the last matrix is H_{xx} . Two points deserve emphasis. First, its diagonal is negative: since f contains $-\gamma_i u_i^2/x_i$, we have $\partial^2 f / \partial x_i^2 = -2\gamma_i u_i^2/x_i^3$, consistent with the concavity of the profit in the state. Second, H_{xx} is not diagonal. The interaction terms $-\beta x_1 x_2 p_1$ and $\delta x_1 x_2 p_2$ in $P^\top b$ are bilinear in the state, so they contribute the off-diagonal entry $\partial^2 H / \partial x_1 \partial x_2 = \delta p_2 - \beta p_1$. This term is the only place where the predator-prey coupling enters the second-order adjoint, and dropping it would decouple the two species precisely where their interaction matters.

To account for the added uncertainty introduced by control-dependent diffusion, we adopt a generalized Hamiltonian function G as formulated in [7], which includes a second-order risk adjustment term:

$$G(X, U, P, R) = f(X, U) + P^\top b(X, U) + \frac{1}{2} \text{tr} \left[\Sigma(X, U)^\top R \Sigma(X, U) \right], \quad (X, U, P, R) \in \mathbb{R}^2 \times \mathcal{U} \times \mathbb{R}^2 \times \mathcal{S}^2, \quad (27)$$

where $\mathcal{U}$ is the set of admissible controls and $\mathcal{S}^2$ is the set of 2×2 symmetric matrices.

Unlike the SMP used in Model A, the stochastic maximum principle for systems with control-dependent diffusion does not maximize G directly. Instead, it guarantees pairs of adjoint processes that maximize an auxiliary function $\mathcal{H}$, defined as:

$$\mathcal{H}(X, U) = G(X, U, P, R) + \text{tr} \left[\Sigma(X, U)^\top (Q - R\Sigma(X, U)) \right], \quad (28)$$

which is associated with an optimal 6-tuple (X^*, U^*, P, Q, R, S) .

Expanding, we obtain

$$\mathcal{H}(X, U) = f(X, U) + P^\top b(X, U) + \frac{1}{2} \text{tr} \left[\Sigma^\top R \Sigma \right] + \text{tr} \left[\Sigma^\top Q - \Sigma^\top R \Sigma \right] \quad (29)$$

$$= f(X, U) + P^\top b(X, U) + \text{tr} \left[\Sigma^\top Q \right] - \frac{1}{2} \text{tr} \left[\Sigma^\top R \Sigma \right]. \quad (30)$$

We now compute $\mathcal{H}(X, U)$ explicitly. Recall:

$$f(X, U) = F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2},$$

$$b(X, U) = \begin{bmatrix} \alpha x_1 - \beta x_1 x_2 - u_1 \\ -\gamma x_2 + \delta x_1 x_2 - u_2 \end{bmatrix}, \quad P = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}, \quad Q = \begin{bmatrix} q_1 & 0 \\ 0 & q_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_{10} x_1 + \sigma_{11} u_1 & 0 \\ 0 & \sigma_{20} x_2 + \sigma_{22} u_2 \end{bmatrix}.$$

Substituting these into $\mathcal{H}$, we obtain:

$$\begin{aligned} \mathcal{H}(X, U) = & F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2} \\ & + p_1(\alpha x_1 - \beta x_1 x_2 - u_1) + p_2(-\gamma x_2 + \delta x_1 x_2 - u_2) \\ & + q_1(\sigma_{10} x_1 + \sigma_{11} u_1) + q_2(\sigma_{20} x_2 + \sigma_{22} u_2) \\ & - \frac{1}{2} [r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1)^2 + r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2)^2]. \end{aligned} \quad (31)$$

To find the optimal control, we compute the first-order conditions by differentiating the Hamiltonian $\mathcal{H}(X, U)$ with respect to each control u_i and setting the result to zero:

$$\frac{\partial \mathcal{H}}{\partial u_1} = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 + q_1 \sigma_{11} - r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1) \sigma_{11}, \quad (32)$$

$$\frac{\partial \mathcal{H}}{\partial u_2} = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 + q_2 \sigma_{22} - r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2) \sigma_{22}. \quad (33)$$

Setting these derivatives equal to zero yields the necessary optimality conditions:

$$0 = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 + q_1 \sigma_{11} - r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1) \sigma_{11}, \quad (34)$$

$$0 = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 + q_2 \sigma_{22} - r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2) \sigma_{22}. \quad (35)$$

As expected, the optimal controls for this model are much more complex than the control-independent diffusion model. Analytic solutions are almost sure to be impossible to find, so we proceed with a numerical approach.

4 Numerical Solution via Forward-Backward Sweep

There are a few approaches to solving FBSDEs numerically [1, 3]. Here, a numerical solution of the stochastic optimal control problem for the predator-prey fishery models is achieved using the *forward-backward sweep method*, an iterative technique designed to solve coupled FBSDEs. This method is well-suited for optimal control problems governed by SDEs, where the state dynamics evolve forward in time and the adjoint processes, derived earlier, evolve backward. The forward-backward sweep method alternates between computing the state trajectories forward in time, solving the adjoint equations backward, and updating the control variables to satisfy the optimality conditions. By iterating these steps, the method converges to the optimal control strategy $(u_1^*(t), u_2^*(t))$, accounting for the stochastic nature of the system through Monte Carlo simulations and ensuring adaptedness to the filtration $\mathcal{F}_t = \sigma(W_1(s), W_2(s) : s \leq t)$. Because of the high degrees of nonlinearity and sensitivity of the models, a weighted update of the controls during the sweep is performed, as in [4, 5], adding a convex combination of the new control values and previous values.

4.1 Model Equations

The system parameters are: intrinsic growth rate of prey $\alpha = 0.5$, predation rate $\beta = 0.02$, natural mortality rate of predator $\gamma = 0.4$, conversion efficiency $\delta = 0.01$, diffusion coefficients $\sigma_1 = \sigma_2 = 0.1$ (Model A), $\sigma_{10} = \sigma_{20} = 0.1$, $\sigma_{11} = \sigma_{22} = 0.05$ (Model B), harvesting profits $F_1 = 10$, $F_2 = 20$, and cost coefficients $\gamma_1 = 50$, $\gamma_2 = 100$. We initialize at the deterministic coexistence equilibrium

$$x_1(0) = \gamma/\delta = 40, \quad x_2(0) = \alpha/\beta = 25,$$

so that all subsequent motion is attributable to harvesting and environmental noise rather than to a transient from an arbitrary starting point. The time horizon is $T = 20$ days.

On the choice of horizon. The horizon deserves comment, because it is constrained by the model rather than chosen freely. The deterministic Lotka–Volterra system conserves the quantity $\delta x_1 - \gamma \log x_1 + \beta x_2 - \alpha \log x_2$, so its orbits are closed and its coexistence equilibrium is a neutral center rather than an attractor. Multiplicative noise destroys this balance: the conserved quantity performs a random walk, the orbits drift outward, and the populations come arbitrarily close to extinction as the horizon grows. Since the harvesting cost $\gamma_i u_i^2/x_i$ diverges as $x_i \rightarrow 0$, the control problem itself degenerates there, and the forward–backward sweep fails to converge because the adjoint forcing $\gamma_i u_i^2/x_i^2$ blows up. Numerically, the minimum uncontrolled population over the simulated paths falls from 4.7 at $T = 20$ to 0.93 at $T = 50$ and 8.3×10^{-3} at $T = 100$; Figure 5 traces this decay. With a Lotka–Volterra period of $2\pi/\sqrt{\alpha\gamma} \approx 14$ days, $T = 20$ covers roughly 1.4 predator–prey cycles and is the longest horizon over which the problem remains well posed at this noise level. Extending the analysis to a full fishing season would require a model with a genuinely stable interior equilibrium — for instance by adding logistic self-limitation $-\alpha x_1^2/K$ to the prey — which we leave to future work.

Recall the forward SDEs governing the state dynamics:

- **Model A (Control-Independent Diffusion):**

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)] dt + \sigma_1 x_1(t) dW_1(t), \quad (36)$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)] dt + \sigma_2 x_2(t) dW_2(t). \quad (37)$$

• **Model B (Control-Dependent Diffusion):**

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)] dt + (\sigma_{10}x_1(t) + \sigma_{11}u_1(t)) dW_1(t), \quad (38)$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)] dt + (\sigma_{20}x_2(t) + \sigma_{22}u_2(t)) dW_2(t). \quad (39)$$

The first-order adjoint BSDEs, derived from the SMP, are:

• **Model A:**

$$\begin{aligned} dp_1(t) &= - \left[(\alpha - \beta x_2(t))p_1(t) + \delta x_2(t)p_2(t) + \sigma_{10}q_1(t) - \frac{\gamma_1 u_1(t)^2}{x_1(t)^2} \right] dt + q_1(t) dW_1(t), \quad (40) \\ dp_2(t) &= - \left[-\beta x_1(t)p_1(t) + (-\gamma + \delta x_1(t))p_2(t) + \sigma_{20}q_2(t) - \frac{\gamma_2 u_2(t)^2}{x_2(t)^2} \right] dt + q_2(t) dW_2(t), \end{aligned} \quad (41)$$

with terminal conditions $p_1(T) = p_2(T) = 0$.

• **Model B:**

$$\begin{aligned} dp_1(t) &= - \left[(\alpha - \beta x_2(t))p_1(t) + \delta x_2(t)p_2(t) + \sigma_{10}q_1(t) - \frac{\gamma_1 u_1(t)^2}{x_1(t)^2} \right] dt + q_1(t) dW_1(t), \quad (42) \\ dp_2(t) &= - \left[-\beta x_1(t)p_1(t) + (-\gamma + \delta x_1(t))p_2(t) + \sigma_{20}q_2(t) - \frac{\gamma_2 u_2(t)^2}{x_2(t)^2} \right] dt + q_2(t) dW_2(t), \end{aligned} \quad (43)$$

with terminal conditions $p_1(T) = p_2(T) = 0$.

For Model B, the second-order adjoint BSDE for the matrix-valued process $R(t) = \begin{bmatrix} r_{11}(t) & r_{12}(t) \\ r_{21}(t) & r_{22}(t) \end{bmatrix}$ is:

$$dR(t) = - \left[b_x^\top R(t) + R(t)b_x + \sum_{j=1}^2 \sigma_x^{j\top} R(t) \sigma_x^j + \sum_{j=1}^2 \left(\sigma_x^{j\top} S_j(t) + S_j(t) \sigma_x^j \right) + H_{xx} \right] dt + \sum_{j=1}^2 S_j(t) dW_j(t), \quad (44)$$

where:

$$\begin{aligned} b_x &= \begin{bmatrix} \alpha - \beta x_2 & -\beta x_1 \\ \delta x_2 & -\gamma + \delta x_1 \end{bmatrix}, \quad \sigma_x^1 = \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix}, \quad \sigma_x^2 = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix}, \\ H_{xx} &= \begin{bmatrix} -2\gamma_1 \frac{u_1^2}{x_1^3} & \delta p_2 - \beta p_1 \\ \delta p_2 - \beta p_1 & -2\gamma_2 \frac{u_2^2}{x_2^3} \end{bmatrix}, \end{aligned}$$

and $S_j(t) = \begin{bmatrix} s_{11}^{(j)}(t) & s_{12}^{(j)}(t) \\ s_{21}^{(j)}(t) & s_{22}^{(j)}(t) \end{bmatrix}$, with terminal condition $R(T) = 0$. Writing out $b_x^\top R + R b_x$ entrywise

and using the symmetry $r_{12} = r_{21}$, the components are

$$dr_{11}(t) = - \left[2(\alpha - \beta x_2)r_{11} + 2\delta x_2 r_{12} + \sigma_{10}^2 r_{11} + 2\sigma_{10}s_{11}^{(1)} - 2\gamma_1 \frac{u_1^2}{x_1^3} \right] dt + s_{11}^{(1)} dW_1(t) + s_{11}^{(2)} dW_2(t), \quad (45)$$

$$dr_{22}(t) = - \left[2(-\gamma + \delta x_1)r_{22} - 2\beta x_1 r_{12} + \sigma_{20}^2 r_{22} + 2\sigma_{20}s_{22}^{(2)} - 2\gamma_2 \frac{u_2^2}{x_2^3} \right] dt + s_{22}^{(1)} dW_1(t) + s_{22}^{(2)} dW_2(t), \quad (46)$$

$$dr_{12}(t) = - \left[(\alpha - \beta x_2 - \gamma + \delta x_1)r_{12} + \delta x_2 r_{22} - \beta x_1 r_{11} + \sigma_{10}s_{12}^{(1)} + \sigma_{20}s_{12}^{(2)} + \delta p_2 - \beta p_1 \right] dt + s_{12}^{(1)} dW_1(t) + s_{12}^{(2)} dW_2(t). \quad (47)$$

The diagonal entries do not form a closed system: r_{11} is forced by r_{12} and vice versa, so (47) must be solved alongside them. This is a consequence of the predator–prey coupling, which enters both through b_x and through the off-diagonal entry of H_{xx} .

The optimal controls are derived from the SMP. For Model A:

$$u_1^*(t) = \max \left(0, \frac{x_1(t)(F_1 - p_1(t))}{2\gamma_1} \right), \quad (48)$$

$$u_2^*(t) = \max \left(0, \frac{x_2(t)(F_2 - p_2(t))}{2\gamma_2} \right). \quad (49)$$

For Model B, the controls satisfy quadratic equations involving $p_j(t)$, $r_{jj}(t)$, and $s_{jj}^{(j)}(t)$, which are solved numerically at each time step to ensure optimality.

4.2 Numerical Algorithms

4.2.1 Overview

The time interval $[0, T]$ is discretized into N points with step size $\Delta t = T/N$. Monte Carlo simulations are performed over M sample paths, with Wiener increments approximated as $\Delta W_j(t_i) \approx \sqrt{\Delta t} \xi_{j,i}^{(m)}$, where $\xi_{j,i}^{(m)} \sim \mathcal{N}(0, 1)$ and are generated once and reused throughout the forward–backward iterations.

Given an initial guess for the control processes, the state equations are simulated forward in time along each Monte Carlo path using the Euler–Maruyama scheme. With the resulting state trajectories fixed, the associated adjoint equations are solved backward in time. The backward dynamics involve conditional expectations that are not available in closed form and are therefore approximated numerically using least-squares regression onto a finite-dimensional space of polynomial basis functions of the state variables. These regression-based estimates provide the stochastic coefficients required to advance the backward equations. The controls are then updated using the resulting adjoint variables, and the procedure is iterated in a forward–backward sweep until convergence.

4.2.2 Model A Algorithms

Algorithm 1 Euler–Maruyama for $x_1(t), x_2(t)$ (Model A)

Require: $x_1(0) = x_{1,0}$, $x_2(0) = x_{2,0}$; controls $u_1(i, m), u_2(i, m)$ for $i = 1, \dots, N$; Δt ; paths M ; steps N ; fixed Gaussians $\xi_1(i, m), \xi_2(i, m) \sim \mathcal{N}(0, 1)$; parameters $\alpha, \beta, \gamma, \delta, \sigma_1, \sigma_2, \varepsilon$.

Ensure: State arrays $x_1(i, m), x_2(i, m)$ for $i = 1, \dots, N+1$.

1: Initialize $x_1(1, m) = x_{1,0}$, $x_2(1, m) = x_{2,0}$ for all $m = 1, \dots, M$.

2: **for** $m = 1$ to M **do**

3: **for** $i = 1$ to N **do**

4: $x_{1,i} \leftarrow x_1(i, m)$, $x_{2,i} \leftarrow x_2(i, m)$.

5: Drift:

$$f_1 = \alpha x_{1,i} - \beta x_{1,i} x_{2,i} - u_1(i, m), \quad f_2 = -\gamma x_{2,i} + \delta x_{1,i} x_{2,i} - u_2(i, m).$$

6: Diffusion:

$$g_1 = \sigma_1 x_{1,i}, \quad g_2 = \sigma_2 x_{2,i}.$$

7: Euler–Maruyama update (uses fixed ξ at (i, m)):

$$x_1(i+1, m) = \max\left(x_{1,i} + f_1 \Delta t + g_1 \sqrt{\Delta t} \xi_1(i, m), \varepsilon\right),$$

$$x_2(i+1, m) = \max\left(x_{2,i} + f_2 \Delta t + g_2 \sqrt{\Delta t} \xi_2(i, m), \varepsilon\right).$$

8: **end for**

9: **end for**

10: **return** x_1, x_2 .

Algorithm 2 Least-Squares Monte Carlo for $p_1(t), p_2(t)$ (Model A)

Require: Forward states $x_1(i, m), x_2(i, m)$; controls $u_1(i, m), u_2(i, m)$; Δt , N , M ; parameters $\alpha, \beta, \gamma, \delta, \sigma_1, \sigma_2, \gamma_1, \gamma_2, \varepsilon$; ridge λ .

Ensure: $p_1(i, m), p_2(i, m)$ for $i = 1, \dots, N+1$ and $q_1(i, m), q_2(i, m)$ for $i = 1, \dots, N$.

1: Initialize $p_1(N+1, m) = 0, p_2(N+1, m) = 0$ for all m .

2: **for** $i = N$ down to 1 **do**

3: Standardize the state, $z_k(m) = (x_k(i, m) - \bar{x}_k)/\text{sd}(x_k)$, and form the quadratic basis

$$\Phi(m, :) = [1, z_1, z_2, z_1^2, z_2^2, z_1 z_2], \quad A = \Phi^\top \Phi + \lambda M \text{diag}(0, 1, 1, 1, 1, 1).$$

4: Regress the next adjoint value on the current state:

$$w_k = A^{-1} \Phi^\top p_k(i+1, :)^T, \quad \mathbb{E}_i[p_k] = (\Phi w_k)^\top, \quad k = 1, 2.$$

5: Recover the martingale coefficient from the Itô representation $p_k = v_k(t, X)$, $q_k = \sigma_k x_k \partial v_k / \partial x_k$:

$$q_1(i, :) = \sigma_1 x_1(i, :) \odot (D_1 w_1)^\top, \quad q_2(i, :) = \sigma_2 x_2(i, :) \odot (D_2 w_2)^\top,$$

where D_k is the design matrix of $\partial \Phi / \partial x_k$.

6: Evaluate the adjoint driver H_x :

$$f_1 = (\alpha - \beta x_2(i, :)) \mathbb{E}_i[p_1] + \delta x_2(i, :) \mathbb{E}_i[p_2] + \sigma_1 q_1(i, :) + \gamma_1 \frac{u_1(i, :)^2}{\max(x_1(i, :)^2, \varepsilon)},$$

$$f_2 = -\beta x_1(i, :) \mathbb{E}_i[p_1] + (-\gamma + \delta x_1(i, :)) \mathbb{E}_i[p_2] + \sigma_2 q_2(i, :) + \gamma_2 \frac{u_2(i, :)^2}{\max(x_2(i, :)^2, \varepsilon)}.$$

7: Step backwards:

$$p_1(i, :) = \mathbb{E}_i[p_1] + f_1 \Delta t, \quad p_2(i, :) = \mathbb{E}_i[p_2] + f_2 \Delta t.$$

8: **end for**

9: **return** p_1, p_2, q_1, q_2 .

Two features of Algorithm 2 are worth stressing, since both are easy to get wrong. First, there is no pathwise $+q\sqrt{\Delta t}\xi$ increment in the update. Taking the conditional expectation $\mathbb{E}_i[\cdot]$ is precisely what removes the martingale term; adding it back would leave p_i depending on the innovation over $[t_i, t_{i+1}]$ and hence not $\mathcal{F}_i$ -measurable, violating the adaptedness the maximum principle requires. Because p_i is returned as $\Phi(x_i)w$, adaptedness holds here by construction.

Second, the martingale coefficient is obtained by differentiating the fitted regression rather than by the textbook estimator $q_i = \mathbb{E}[p_{i+1}\Delta W_i \mid \mathcal{F}_i]/\Delta t$. The two are equivalent in the limit, but the latter divides a sample covariance by Δt and therefore has variance of order $1/\Delta t$, so it degrades as the grid is refined — exactly the opposite of what a discretization should do. On the validation problem of Section 4.2.3 the derivative form is roughly 30 times more accurate at the same sample size (1.6% versus 52% relative error), and unlike the covariance estimator it requires no access to the driving noise at all.

Algorithm 3 Forward–Backward Sweep Method (Model A)

Require: $T, N, M, \Delta t = T/N$; tolerance `tol`; max iterations K ; bound $u_{\max}$; relaxation $\omega \in (0, 1)$; parameters; initial states $x_{1,0}, x_{2,0}$.

Ensure: Approx. optimal controls u_1, u_2 and associated state/adjoint trajectories.

- 1: Initialize $u_1(i, m) = 0, u_2(i, m) = 0$ for $i = 1, \dots, N$.
- 2: Generate **once**: $\xi_1(i, m), \xi_2(i, m) \sim \mathcal{N}(0, 1)$ for all $i = 1, \dots, N, m = 1, \dots, M$.
- 3: **for** $k = 1$ to K **do**
- 4: Forward: compute (x_1, x_2) using Algorithm 1 with fixed ξ .
- 5: Backward: compute (p_1, p_2, q_1, q_2) using Algorithm 2.
- 6: Compute the pointwise Hamiltonian maximizer:

$$\tilde{u}_1(i, m) = \min \left(u_{\max}, \max \left(0, \frac{x_1(i, m)(F_1 - p_1(i, m))}{2\gamma_1} \right) \right),$$

$$\tilde{u}_2(i, m) = \min \left(u_{\max}, \max \left(0, \frac{x_2(i, m)(F_2 - p_2(i, m))}{2\gamma_2} \right) \right).$$

- 7: Compute the undamped fixed-point residual

$$\text{res} = \max \left(\max_{i,m} |\tilde{u}_1 - u_1|, \max_{i,m} |\tilde{u}_2 - u_2| \right).$$

- 8: **if** `res` < `tol` **then**
- 9: $u_1 \leftarrow \tilde{u}_1, u_2 \leftarrow \tilde{u}_2$; **break**
- 10: **end if**
- 11: Relaxation / weighted update:

$$u_1 \leftarrow \omega u_1 + (1 - \omega) \tilde{u}_1, \quad u_2 \leftarrow \omega u_2 + (1 - \omega) \tilde{u}_2.$$

- 12: **end for**

- 13: **return** u_1, u_2 and the latest (x_1, x_2, p_1, p_2) .
-

The residual in Algorithm 3 is measured on the undamped update $\tilde{u}$, not on the relaxed step $u^{new} - u$. This distinction is not cosmetic: since $u^{new} - u = (1 - \omega)(\tilde{u} - u)$, testing the step against a tolerance `tol` is equivalent to testing the true residual against `tol`/(1 - ω), which at $\omega = 0.9$ is ten times looser than intended. A sweep that is in fact still far from its fixed point — or, worse, one that is about to diverge — can then be reported as converged.

4.2.3 Implementation and Validation

The forward dynamics are discretized using the Euler–Maruyama scheme. The adjoint equations use an explicit backward step: the driver is evaluated at the projected time- i values, giving a scheme that is first-order accurate in Δt and conditionally — not unconditionally — stable, so the grid must be fine enough for the horizon considered. To prevent numerical degeneracy, a small threshold $\epsilon > 0$ is introduced in denominators involving the state variables; we report the minimum population attained and verify that this floor never binds, since a binding floor signals that the model has left its domain of validity. All reported results use $N = 2000$ time steps, $M = 1000$ Monte Carlo paths, damping $\omega = 0.9$ and tolerance 10^{-3} .

Because the correctness of a forward–backward sweep is not self-evident from its output — a

plausible-looking control trajectory can be produced by an adjoint with a sign error — we validate the implementation against independently known quantities rather than against its own internal consistency. Four checks are used.

- (i) **Forward solver.** Setting $\beta = \delta = 0$ and $u \equiv 0$ decouples the system into geometric Brownian motions with $\mathbb{E}[x_1(t)] = x_1(0)e^{\alpha t}$ and $\mathbb{E}[x_2(t)] = x_2(0)e^{-\gamma t}$. The simulated means agree with these to within one Monte Carlo standard error.
- (ii) **Adjoint equations.** In the deterministic limit the exact sensitivity of the objective to a perturbation of u_1 held over a single step is $\partial J / \partial u_1(t_i) = (F_1 - 2\gamma_1 u_1 / x_1 - p_1) \Delta t$. Comparing this against a central finite difference of J , evaluated at a deliberately sub-optimal control so that both sides are far from zero, tests every sign in b , f and H_x simultaneously. The agreement, plotted in the left panel of Figure 4, is 8.3×10^{-3} relative error at $\Delta t = 10^{-2}$, halving to 4.1×10^{-3} , 2.0×10^{-3} and 1.0×10^{-3} as the step is successively halved. The clean first-order decay identifies the discrepancy as discretization error rather than a modeling or sign error; a sign error would produce an $O(1)$ mismatch that does not vanish with Δt .
- (iii) **BSDE recursion.** For $dx = \mu x dt + \sigma x dW$, the BSDE $dp = -[-\mu x + (q - \sigma x)]dt + q dW$ with $p(T) = x(T)$ has the exact solution $(p, q) = (x, \sigma x)$. Since the driver depends on q with an $O(1)$ coefficient, an inaccurate q propagates directly into p . The scheme recovers $p(0)$ to 0.4% and q to 1.6%.
- (iv) **Optimality of the computed control.** At the fixed point we report the Karush–Kuhn–Tucker residual for $\max_u H$ subject to $u \geq 0$ — that is, $|\partial H / \partial u_i|$ at interior points and $\max(0, \partial H / \partial u_i)$ where the non-negativity constraint is active, since $\partial H / \partial u_i$ need not vanish on the boundary. We additionally perturb the computed control by 200 independent random walks and re-evaluate J along the same Brownian paths; the resulting shortfalls are shown in the right panel of Figure 4.

We note that verifying the martingale property by checking that empirical averages of $p_j(t_i, m) \xi_{j,i}^{(m)}$ vanish, as is sometimes suggested, is not informative for this scheme. Under Algorithm 2 the adjoint is returned as a deterministic function of the time- i state, so this correlation is zero by construction; and it is also close to zero for a scheme in which q is erroneously identically zero. The check can be passed by a solver that is badly wrong.

4.2.4 Model B Algorithms

5 Simulation Results

5.1 Model A

With the parameters of Section 4.2.3, the forward–backward sweep converges monotonically, reaching a residual of 9.9×10^{-4} after 233 iterations; as Figure 3 shows, the residual decays linearly on a logarithmic scale, indicating a geometric rate of contraction. At the fixed point the KKT residual is at most 1.0×10^{-3} for u_1 and 2.2×10^{-3} for u_2 , comparable to the sweep tolerance itself. The non-negativity constraint is nowhere active and the safeguard $u_{\max}$ never binds, so the computed control is an interior stationary point of the Hamiltonian; since $\partial^2 H / \partial u_i^2 = -2\gamma_i / x_i < 0$, it is a maximum. The minimum population attained across all paths and times is $x_1 = 5.44$, $x_2 = 3.46$, comfortably away from the positivity floor.

The optimal policy yields $J = 881.2$. Of 200 random-walk perturbations of the computed control, re-simulated along identical Brownian paths, none improved the objective; the best attained 880.2. Figure 1 shows the resulting trajectories. Mean harvest rates are $\bar{u}_1 = 4.49$ and $\bar{u}_2 = 2.18$, and the populations complete approximately 1.4 Lotka–Volterra cycles over the horizon, with $\mathbb{E}[x(T)] = (43.4, 19.1)$ against $x(0) = (40, 25)$.

The economically interesting feature is the sign of the shadow prices, shown in Figure 2. At $t = 0$ we find $\mathbb{E}[p_1(0)] = 2.82 > 0$ but $\mathbb{E}[p_2(0)] = -1.72 < 0$, and this ordering is typical rather than universal: over the horizon p_1 is positive on 76% of the time nodes and p_2 negative on 64%, with time averages of $+1.11$ and -1.83 respectively. Both processes oscillate with the predator–prey cycle and change sign as the cycle turns, so the interpretation below describes the prevailing regime, not a pointwise property.

The prey carries a positive *in situ* value. Because $u_1^* = x_1(F_1 - p_1)/(2\gamma_1)$, a positive p_1 depresses harvesting below the myopic rate $x_1 F_1/(2\gamma_1)$, the shadow price acting as a conservation premium on a stock worth more left in the water than landed immediately. The predator’s shadow price is predominantly negative, so over most of the horizon the optimal policy harvests it more aggressively than the myopic rule. This is not a pricing effect — the predator commands the higher unit price, $F_2 = 20$ against $F_1 = 10$ — but an ecological one: each predator suppresses the prey stock, and the resulting loss of future prey harvest outweighs the predator’s own marginal value. The optimal strategy thus uses harvesting partly as a mechanism of predator control, a conclusion that is invisible to any single-species treatment and that emerges here directly from the coupling term $\delta x_2 p_2$ in the adjoint for p_1 .

6 Discussion

7 Conclusion and Future Work

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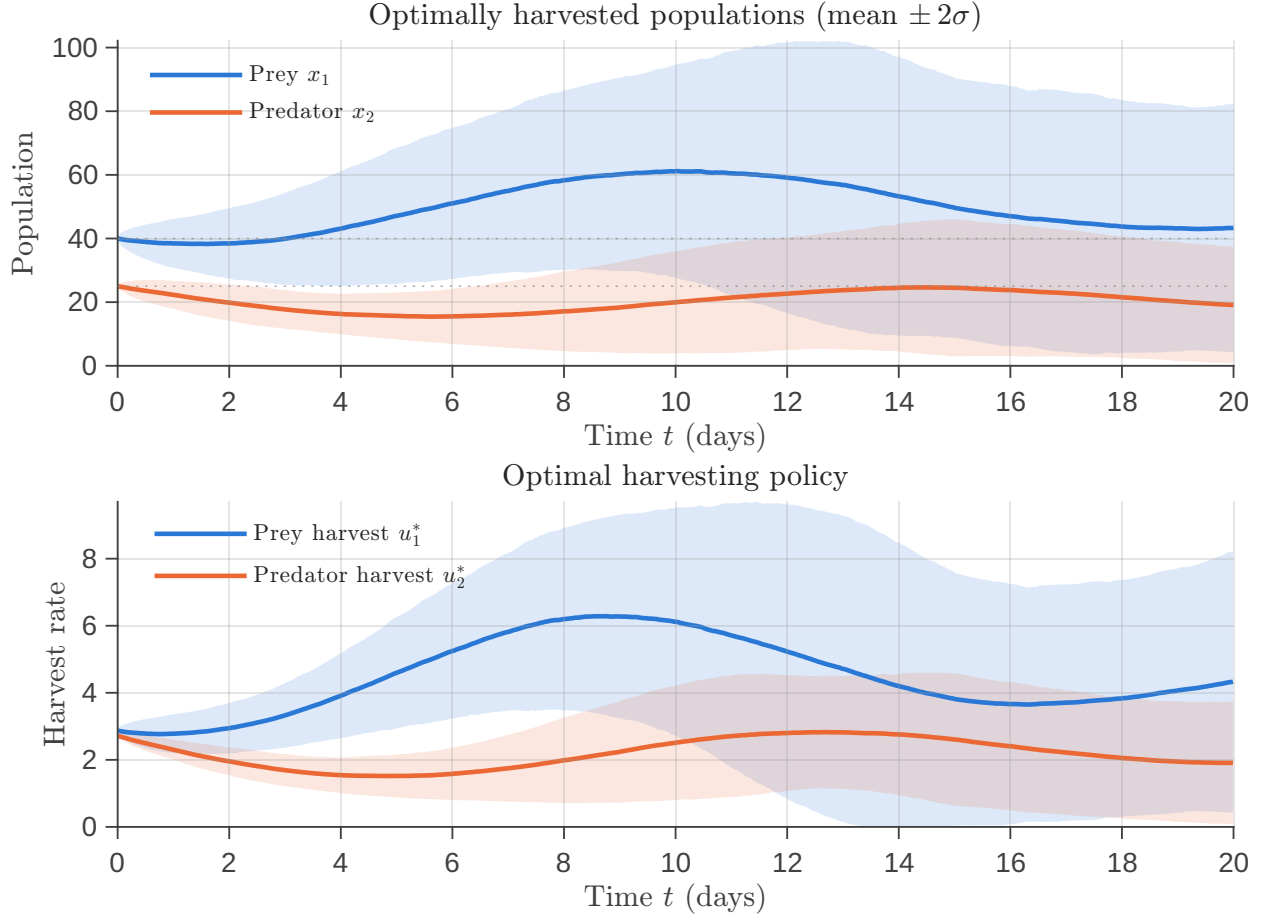


Figure 1: Optimally harvested populations and the corresponding harvesting policy, as means over $M = 1000$ Monte Carlo paths with $\pm 2\sigma$ bands. Dotted lines in the upper panel mark the deterministic coexistence equilibrium $(\gamma/\delta, \alpha/\beta) = (40, 25)$ from which the paths are started. Both harvest rates track their own stock, and the populations complete roughly 1.4 predator–prey cycles over the 20-day horizon.

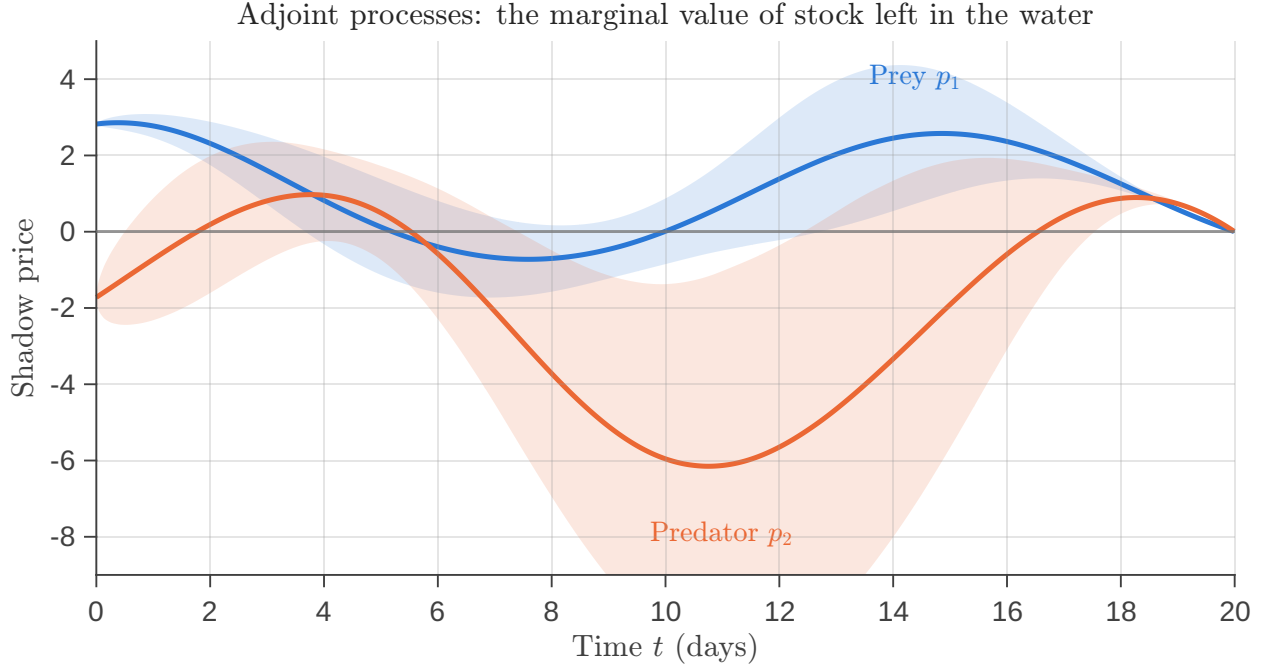


Figure 2: The adjoint processes, interpretable as the marginal value of a unit of stock left unharvested. The prey shadow price is predominantly positive, acting as a conservation premium that holds harvesting below the myopic rate; the predator shadow price is predominantly negative, because a predator’s suppression of the prey stock outweighs its own market value. Both oscillate with the predator–prey cycle and reverse sign as it turns.

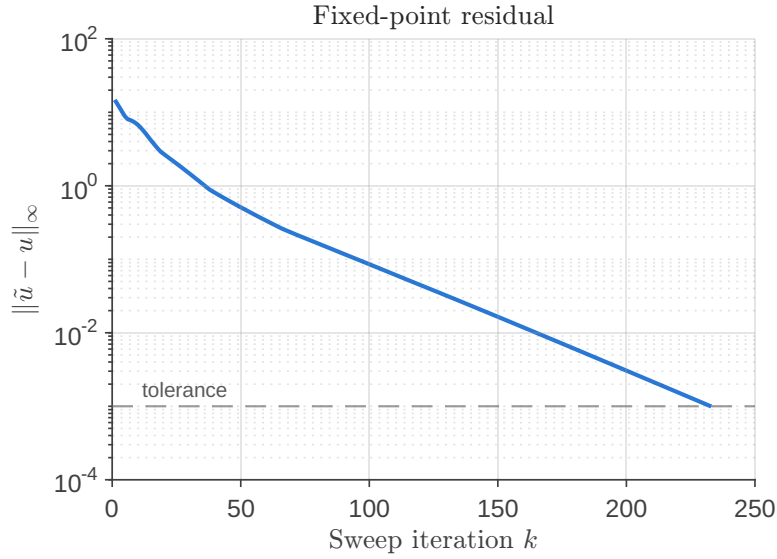


Figure 3: Convergence of the forward–backward sweep. The residual is measured on the undamped update $\tilde{u}$; its linear decay on a logarithmic scale indicates a geometric rate of contraction.

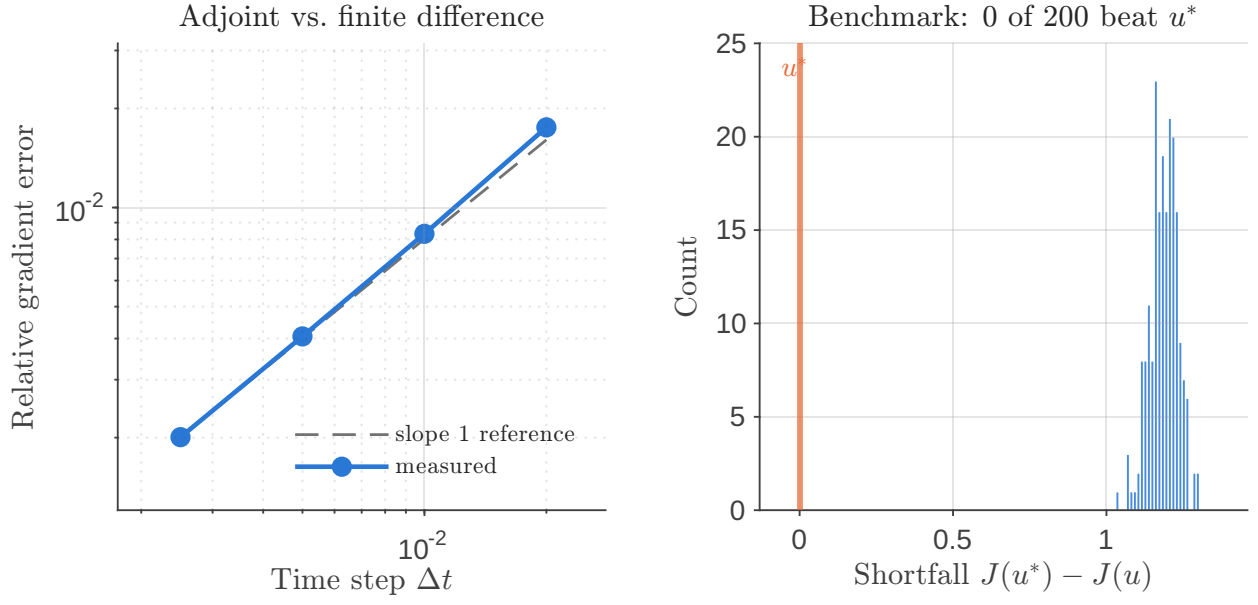


Figure 4: Validation. Left: the adjoint gradient against a central finite difference of J in the deterministic limit, evaluated at a sub-optimal control. The error follows the slope-1 reference, confirming that the discrepancy is first-order discretization error rather than a sign error, which would not vanish with Δt . Right: the shortfall $J(u^*) - J(u)$ over 200 random-walk perturbations of the computed control, re-simulated along identical Brownian paths. Every perturbation falls short.

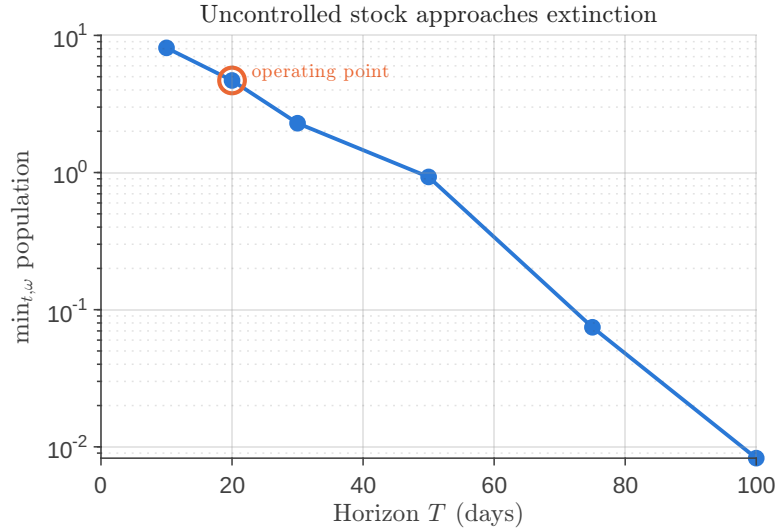


Figure 5: Minimum population attained by the uncontrolled system, over all paths and times, as a function of the horizon. The neutral center of the Lotka–Volterra system offers no restoring force, so multiplicative noise drives the orbits outward and the stock toward extinction. Since the harvesting cost $\gamma_i u_i^2 / x_i$ diverges as $x_i \rightarrow 0$, this bounds the horizon over which the control problem is well posed.